SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	U.UDHAYASHREE	**Reg. No.:**	192521448
Section / Batch:	BTECH-IT	**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
		GRAND TOTAL:	**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

Q1. Recursive Financial Summation Engine

A financial computation system reduces the input size by half and performs linear work at each recursive call to compute cumulative summaries. Apply the substitution method to solve the recurrence $T(n)=T(n/2)+n$. Expand the recurrence step-by-step and verify the final result. Analyze how the algorithm behaves for large-scale transaction data. Interpret the efficiency using asymptotic notation.

Code of Conduct

I U.Udhayashree 192521448 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _________________	Signature: _________________	Signature: _________________
Date: ______________________	Date: ______________________	Date: ______________________

1. Social Media Feed Processing

Problem Statement

A financial system computes cumulative transaction summaries by recursively dividing the input data into two equal halves and performing linear work at each step. The running time is given by the recurrence relation:

$$T(n) = T(n/2) + n$$

Use the **substitu**

tion method to solve the recurrence, determine its time complexity, and analyze its efficiency for large-scale financial transaction data using asymptotic notation.

Answer

Introduction

Recursive algorithms are widely used in financial computing to process large volumes of transaction data efficiently. By dividing the input into smaller subproblems and combining the results, these algorithms reduce computational complexity and improve scalability. The recurrence relation $T(n) = T(n/2) + n$represents a recursive process where the input size is halved at each step while performing linear work to compute cumulative summaries. Solving this recurrence using the substitution method helps determine the algorithm's time complexity and evaluate its efficiency for handling large-scale financial transaction data.

Substitution Method

The substitution method is a mathematical technique used to solve recurrence relations by repeatedly replacing the recursive term with its equivalent expression until the base case is reached. After identifying the expansion pattern, the terms are simplified to obtain the final time complexity. This method is commonly used to analyze the efficiency of recursive algorithms and helps determine their asymptotic behavior..

Given,

T(n)=T(n/2)+n

STEP 1: EXPAND ONCE

Substitute $T(n/2)$:

$$T(n) = \left[T(n/4) + \frac{n}{2}\right] + n$$
$$T(n) = T(n/4) + \frac{n}{2} + n$$

Expand again:

$$T(n) = \left[T(n/8) + \frac{n}{4}\right] + \frac{n}{2} + n$$
$$T(n) = T(n/8) + \frac{n}{4} + \frac{n}{2} + n$$

After **k** expansions:

$$T(n) = T\left(\frac{n}{2^k}\right) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}}\right)$$

STEP 2: REACH THE BASE CASE

The recursion stops when:

$$\frac{n}{2^k} = 1$$

Therefore,

$$2^k = n$$
$$k = \log_2 n$$

Substitute into the recurrence:

$$T(n) = T(1) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{\log n - 1}}\right)$$

The geometric series is:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2\left(1 - \frac{1}{n}\right) < 2$$

Hence,

$$T(n) = 1 + 2n - 2$$
$$T(n) = 2n - 1$$

STEP 3: FINAL RESULT

Ignoring constants in asymptotic analysis,

$$T(n) = \Theta(n)$$

General Expansion

Starting with the recurrence relation:

$$T(n) = T(n/2) + n$$

After repeated substitutions, the recurrence expands as:

$$T(n) = T(n/4) + \frac{n}{2} + n$$
$$T(n) = T(n/8) + \frac{n}{4} + \frac{n}{2} + n$$
$$T(n) = T(n/16) + \frac{n}{8} + \frac{n}{4} + \frac{n}{2} + n$$

After **k** expansions, the general form is:

$$T(n) = T\left(\frac{n}{2^k}\right) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}}\right)$$

When the recursion reaches the base case,

$$\frac{n}{2^k} = 1$$
$$k = \log_2 n$$

Substituting this into the general expansion gives:

$$T(n) = T(1) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{\log_2 n - 1}}\right)$$

This is the **general expansion** of the recurrence using the substitution method.

Analysis for Large-Scale Financial Transaction Data

- The input size is reduced by half at every recursive call.
- At each level, only linear work (n) is performed to compute cumulative financial summaries.
- Although the recursion depth is $\log n$, the total work forms a geometric series that converges to 2n, making the overall running time linear.
- Therefore, even for millions of financial transactions, the execution time grows proportionally to the number of transactions, making the algorithm highly scalable and efficient.

Asymptotic Efficiency

- **Time Complexity:** $\boxed{\Theta(n)}$
- **Best Case:** $O(n)$
- **Average Case:** $O(n)$
- **Worst Case:** $O(n)$

Conclusion

Using the substitution method, the recurrence relation

$$T(n) = T(n/2) + n$$

expands into a geometric series whose total cost is approximately **2n**. Thus, the algorithm has a **linear time complexity,** $\Theta(n)$, making it suitable for efficiently processing large-scale financial transaction data.